

TrendWatch AI — Social Media Intelligence Report

Use Case 2: Brand monitoring on Reddit · Generated 2026-06-09 03:41 UTC

Query: nvidia r/technology Limit: 50 Time: devvit_import

Data: Live Reddit (imported from Devvit app)

Report contents (assignment sections)

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1. Executive briefing

Source: rule_based_fallback · RAG + CoT LLM or rule-based fallback

Executive summary

TrendWatch AI analysed 50 Reddit posts. Negative sentiment is 14.0%. Crisis score is 16/100 (Low).

Main concerns

- Risk keywords appeared 2 times across posts.
- 5 high-engagement posts carry negative sentiment.

Crisis explanation

Risk keywords appeared 2 times across posts. 5 high-engagement posts carry negative sentiment.

Recommended actions

- Monitor high-impact negative threads daily.
- Prepare a factual FAQ for top complaint topics.
- Escalate to customer support if refund/delay keywords spike.

Communication strategy

Address emerging topics: Nvidia / Chip / Windows, Nvidia / With / Huang, Nvidia / Huang / Jensen. Use empathetic, transparent tone; avoid dismissive responses.

Full LLM report (markdown)

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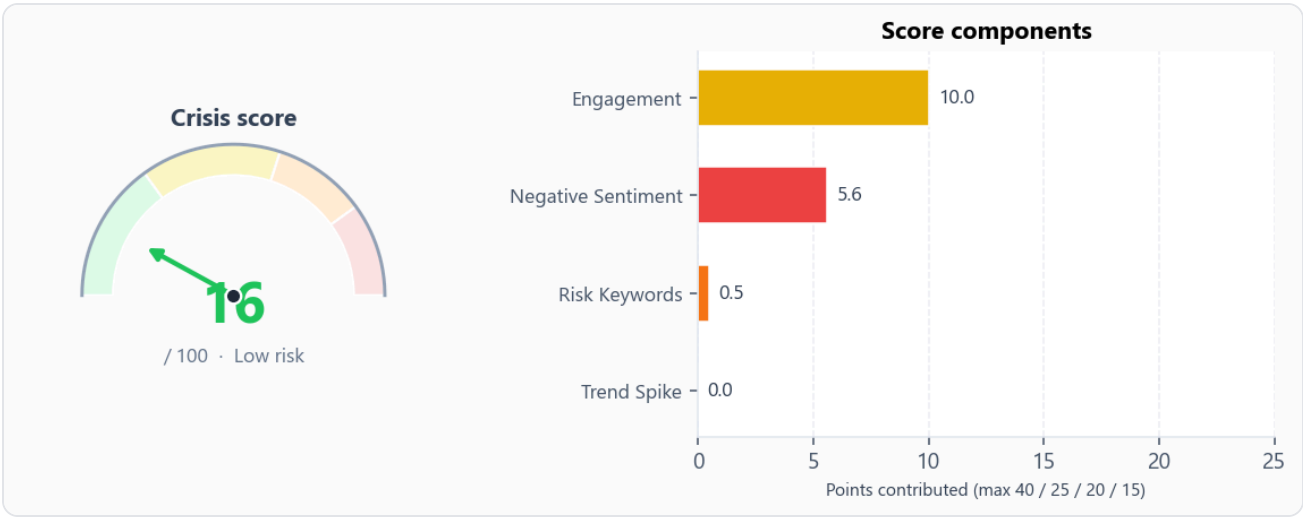
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Ethics notice: Analysis uses public Reddit text. Sentiment and crisis scores are algorithmic estimates, not legal or PR advice. Models may reflect language bias; human review is required before brand actions.

2. Crisis intelligence layer



16/100 — Low risk

Weighted fusion of negative sentiment, risk keywords, high-engagement negative posts, and trend spikes.

Score components

Component	Points
negative_sentiment	5.6
risk_keywords	0.5
engagement	10.0
trend_spike	0.0

Reasons

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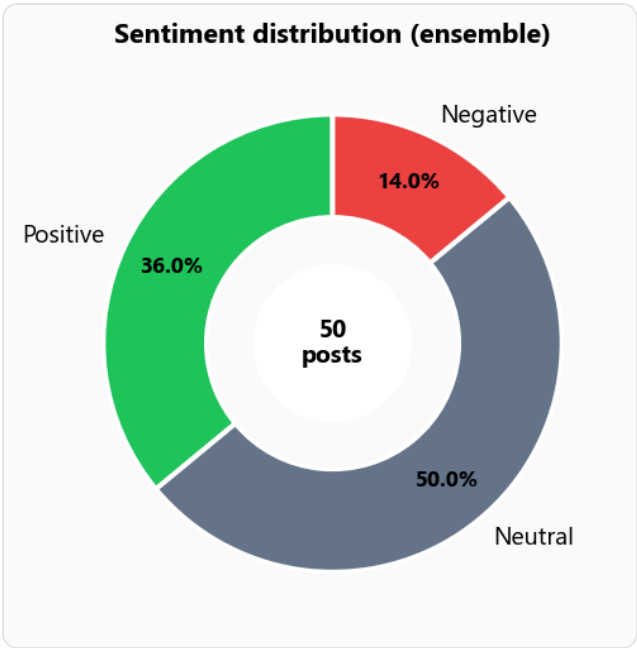
3. Overview & sentiment analysis

50
POSTS ANALYSED

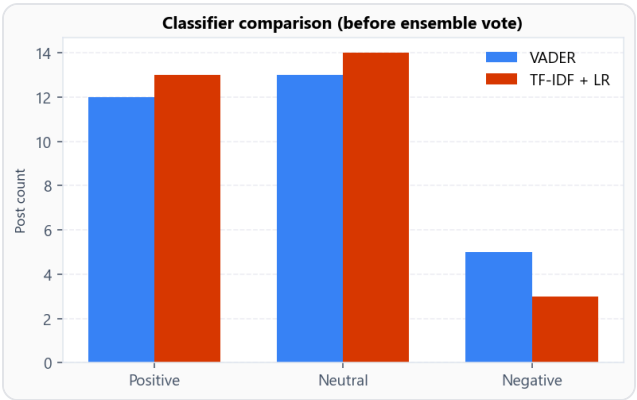
36.0%
POSITIVE

50.0%
NEUTRAL

14.0%
NEGATIVE



Final labels after VADER + LR ensemble majority vote.



Independent classifier outputs before ensemble fusion.

Sentiment distribution (VADER + LR ensemble)

Positive36.0%

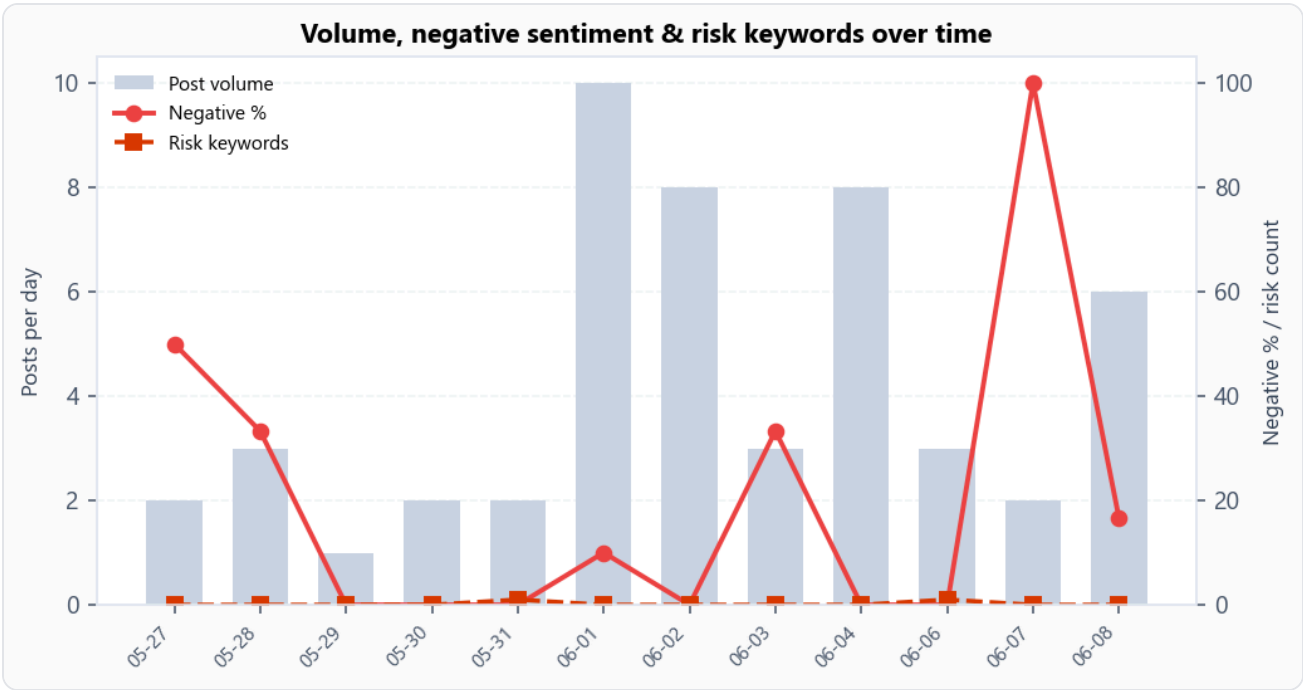
Neutral50.0%

Negative14.0%

Label	Count
positive	18
neutral	25
negative	7

Counts: positive=18, neutral=25, negative=7

4. Trend detection



Daily post volume (bars), negative sentiment % (line), and risk keyword counts (dashed).

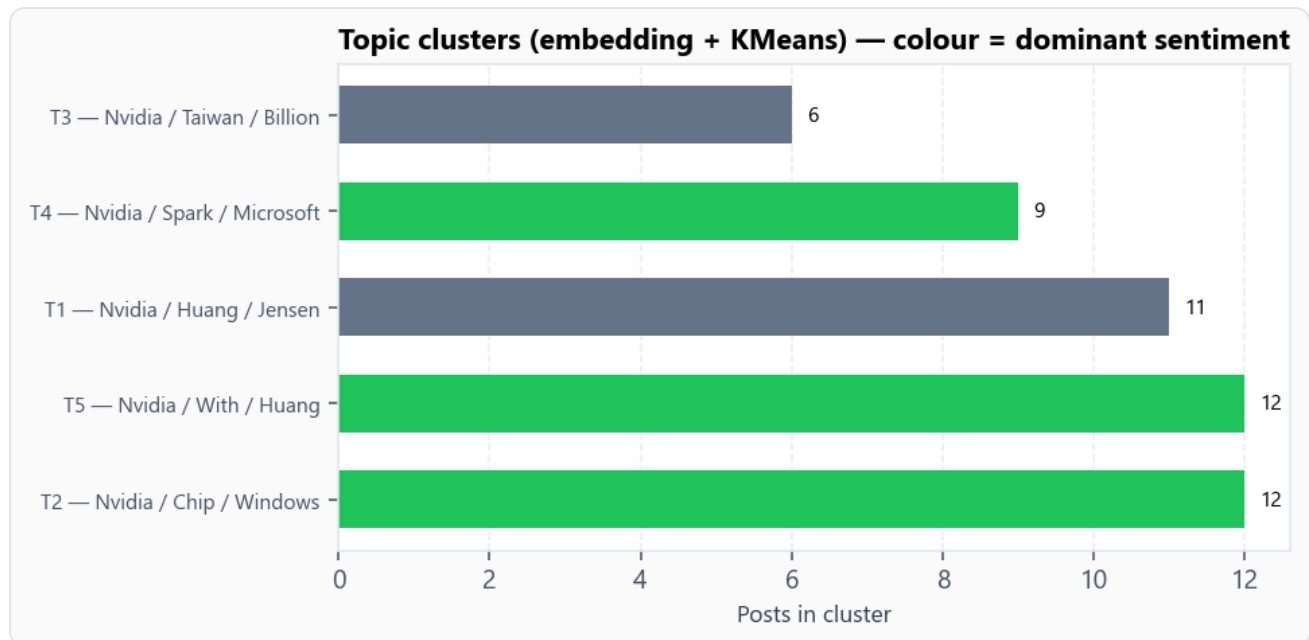
Spike detected: No — No significant negative sentiment spike detected.

Average negative ratio: 20.3%

Date	Posts	Negative	Neg %	Risk keywords
2026-05-27	2	1	50.0%	0
2026-05-28	3	1	33.3%	0
2026-05-29	1	0	0.0%	0
2026-05-30	2	0	0.0%	0
2026-05-31	2	0	0.0%	1
2026-06-01	10	1	10.0%	0
2026-06-02	8	0	0.0%	0
2026-06-03	3	1	33.3%	0
2026-06-04	8	0	0.0%	0
2026-06-06	3	0	0.0%	1

Date	Posts	Negative	Neg %	Risk keywords
2026-06-07	2	2	100.0%	0
2026-06-08	6	1	16.7%	0

5. Topic clusters (embeddings + KMeans)



Topic 1: Nvidia / Chip / Windows

Size: 12 · Avg sentiment: **positive**

Keywords: nvidia, chip, windows, memory, development, planning

Representative: First Windows PCs powered by Nvidia chips to debut next week

Topic 4: Nvidia / With / Huang

Size: 12 · Avg sentiment: **positive**

Keywords: nvidia, with, huang, superchip, into, spark

Representative: Nvidia and Microsoft Researchers Say AI Agents Don't Care About Safety or Reliability

Topic 0: Nvidia / Huang / Jensen

Size: 11 · Avg sentiment: **neutral**

Keywords: nvidia, huang, jensen, chip, from, warren

Representative: 'Just too lazy': Nvidia's Jensen Huang slams CEOs for blaming job losses on AI

Topic 3: Nvidia / Spark / Microsoft

Size: 9 · Avg sentiment: **positive**

Keywords: nvidia, spark, microsoft, surface, laptop, laptops

Representative: Microsoft builds its ultimate MacBook Pro rival with the NVIDIA powered Surface Laptop Ultra

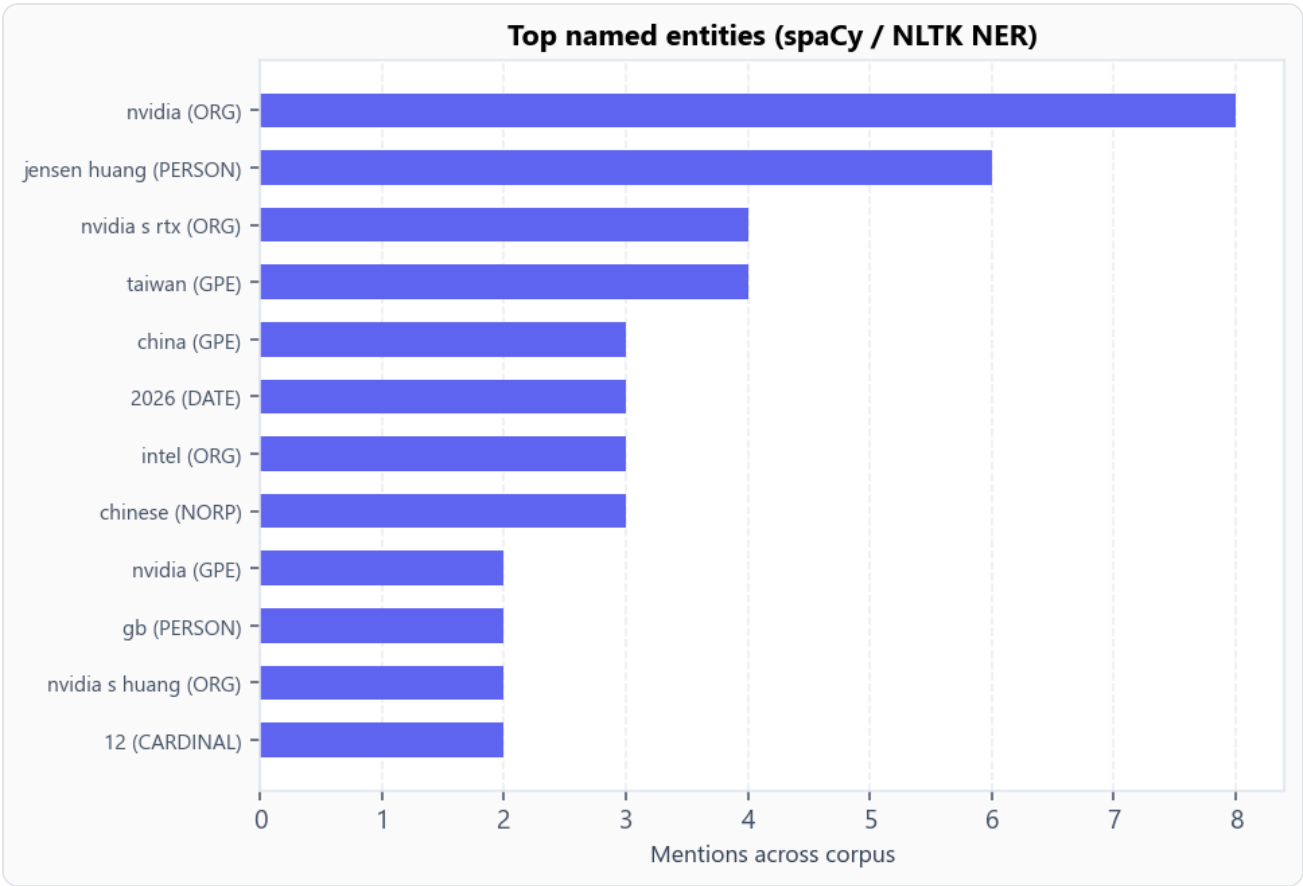
Topic 2: Nvidia / Taiwan / Billion

Size: 6 · Avg sentiment: **neutral**

Keywords: nvidia, taiwan, billion, year, crackdown, hits

Representative: Taiwan just busted an alleged Nvidia GPU smuggling ring using Japan as a cover

6. Named entities (NER)



Entity	Mentions
nvidia (ORG)	8
jensen huang (PERSON)	6
nvidia s rtx (ORG)	4
taiwan (GPE)	4
china (GPE)	3
2026 (DATE)	3
intel (ORG)	3
chinese (NORP)	3
nvidia (GPE)	2
gb (PERSON)	2

Entity	Mentions
nvidia s huang (ORG)	2
12 (CARDINAL)	2
microsoft (ORG)	2
first (ORDINAL)	2
five (CARDINAL)	1

7. Risk keywords & high-impact posts

Risk keyword frequency

Keyword	Count
leak	2

Similar post pairs (cosine similarity)

Post A	Post B	Similarity
Nvidia, SK to detail cooperation plan as Huang flags pr	Nvidia, SK to detail cooperation plan as Jensen Huang f	0.959
CEO Jensen Huang says Nvidia is too busy with the 'giga	Jensen Huang says Nvidia wants to 'reinvent the single	0.829
Nvidia launches 'superchip' putting AI power into lapto	t3_1ttuyvs	0.799
t3_1tq25ci	t3_1tpb2iu	0.796
The Nvidia RTX Spark Era Starts Here: Hands On With Mic	Here's a Look at Every Nvidia RTX Spark Laptop Announce	0.776

High-impact posts

Title	Subreddit	Score	Comments	Sentiment	Risk terms	Impact
'Just too lazy': Nvidia's Jensen Huang slams CEOs for blaming job losses on AI	technology	1757	122	negative	—	12.28

Title	Subreddit	Score	Comments	Sentiment	Risk terms	Impact
Nvidia CEO Jensen Huang declines Warren’s request to testify at AI hearing	technology	1253	73	neutral	—	11.44
Nvidia and Microsoft Researchers Say AI Agents Don't Care About Safety or Reliab...	technology	554	121	positive	—	11.12
Nvidia's Huang calls AI job fears 'nonsense'	technology	319	208	negative	—	11.11
First Windows PCs powered by Nvidia chips to debut next week	technology	353	164	neutral	—	10.98

8. Post-level analysis (sample)

Showing up to 30 posts with VADER label, LR label, and extracted risk terms.

Title	Author	Ensemble	VADER	LR	Risk terms
From barbecue diplomacy to AI deals: Five takeaways from Nvidia CEO Je...	Logical_Welder3467	neutral	neutral	neutral	—
Nvidia CEO Jensen Huang declines Warren's request to testify at AI hea...	Plastic_Ninja_9014	neutral	neutral	neutral	—
Google and Nvidia Consider Intel as Backup Chip Manufacturer	Logical_Welder3467	neutral	neutral	neutral	—
Nvidia and SK hynix ink multi-year memory co-development and supply ag...	Logical_Welder3467	positive	positive	positive	—
HP announces the most powerful Windows AI PC ever built — Nvidia GB300...	Logical_Welder3467	positive	positive	positive	—
Nvidia's Huang calls AI job fears 'nonsense'	IKeepItLayingAround	negative	negative	neutral	—
Nvidia, SK to detail cooperation plan as Huang flags prolonged chip sh...	Logical_Welder3467	negative	negative	negative	—
Nvidia, SK to detail cooperation plan as Jensen Huang flags prolonged ...	boppinmule	negative	negative	negative	—
Nvidia's RTX Spark Laptops Look Hell-Bent on Disruption	Logical_Welder3467	neutral	negative	positive	—
Elizabeth Warren wants Jensen Huang to testify on Nvidia's chip sales ...	Just-Grocery-2229	neutral	neutral	neutral	—

Title	Author	Ensemble	VADER	LR	Risk terms
Nvidia is reportedly still planning fabled RTX 50 Super series for 202...	Logical_Welder3467	positive	positive	positive	leak
Nvidia's RTX Spark laptops are gunning for the MacBook Pro. Yawn	Few_Baseball_3835	positive	positive	positive	—
The Nvidia RTX Spark Era Starts Here: Hands On With Microsoft's Surfac...	Logical_Welder3467	positive	positive	positive	—
Intel and AMD are tumbling after Nvidia muscles into PC chip territory	Frosty-Bit4667	neutral	neutral	neutral	—
Nvidia launches 'superchip' putting AI power into laptops and PCs	Frosty-Bit4667	neutral	neutral	neutral	—
Intel CEO praises 'friend' Nvidia and TSMC despite rising competition	Logical_Welder3467	positive	positive	positive	—
Nvidia GPU crackdown hits China-linked Southeast Asia data centers	Dramatic-Shake-8888	neutral	neutral	neutral	—
CEO Jensen Huang says Nvidia is too busy with the 'gigantic project' o...	Logical_Welder3467	positive	positive	positive	—
Marvell stock soars 32% as Nvidia's Huang says it could be the next tr...	Logical_Welder3467	neutral	neutral	neutral	—
Nvidia is already planning N2X and N3X chips — the goal is the Star Tr...	dapperlemon	neutral	neutral	neutral	—
AMD executives react to Nvidia's RTX Spark — 'you're just wrong if you...	Logical_Welder3467	negative	negative	negative	—

Title	Author	Ensemble	VADER	LR	Risk terms
Nvidia's Big H-1B Push Is Silver Lining For Indian Techies In US	Conscious-Quarter423	neutral	neutral	neutral	—
Here's a Look at Every Nvidia RTX Spark Laptop Announced So Far	Logical_Welder3467	positive	positive	positive	—
Microsoft debuts Surface RTX Spark Dev Box — Nvidia-powered mini-PC he...	Logical_Welder3467	positive	positive	positive	—
Nvidia and Microsoft Researchers Say AI Agents Don't Care About Safety...	MarvelsGrantMan136	positive	positive	positive	—
Jensen Huang says Nvidia wants to 'reinvent the single most important ...	Logical_Welder3467	positive	positive	positive	—
Chinese military has been acquiring Nvidia chips, even post-Washingto...	sr_local	neutral	neutral	neutral	—
Nvidia chases \$200B CPU market with AI agent PCs from Microsoft, Dell,...	Quantum-Coconut	neutral	neutral	neutral	—
Nvidia's RTX Spark could capitalize where Qualcomm's Arm-based effort...	Logical_Welder3467	positive	positive	positive	—
Nvidia picks Unitree for humanoid robot platform as Chinese startup ey...	Logical_Welder3467	neutral	neutral	neutral	—

9. NLP techniques demonstrated (rubric)

Minimum **3 basic** + **3 advanced** techniques — implementation mapping for your written report.

Basic techniques

- **Preprocessing** (tokenize, stem, lemmatize, URL removal)
- **Named entity recognition** (spaCy / NLTK)
- **Part-of-speech tagging**
- **TF-IDF + Logistic Regression classification**
- **Sentiment ensemble** (VADER + LR majority vote)
- **Embedding-based topic clustering** (KMeans)
- **Regex information extraction** (hashtags, mentions, risk lexicon)

Advanced techniques

- **Foundation LLM** (OpenAI-compatible briefing)
- **Retrieval-augmented generation** (top-k posts)
- **Prompt engineering** (structured analyst template)
- **Chain-of-thought prompting**
- **LLM-as-judge / automated evaluation**
- **Ensemble methods** (sentiment + multi-signal crisis)
- **Agentic ReAct-style orchestration**
- **Ethics & fairness disclaimers**

10. Agent orchestration & evaluation

Agent status: completed

ReAct-style trace

Step	Thought	Action	Observation
1	Need Reddit posts about the brand/topic.	collect_posts	Starting data collection.
2	Collected 50 posts.	collect_posts	Received 50 posts.
3	Raw text must be normalized for NLP.	preprocess	Running tokenizer pipeline.
4	Preprocessing complete.	preprocess	50 posts prepared.
5	Extract hashtags, mentions, risk keywords.	extract	Regex IE.
6	Run NER and POS for entity-aware monitoring.	ner_pos	spaCy/NLTK.
7	Classify sentiment with VADER + LR ensemble.	sentiment	Ensemble vote.
8	Cluster topics with embeddings.	topics	KMeans clustering.
9	Compute temporal trends.	trends	Daily aggregation.
10	Score crisis risk from sentiment + engagement + trends.	crisis	Crisis formula.
11	Retrieve relevant posts for grounded LLM briefing (RAG).	rag	Query=nvidia
12	Generate executive briefing with CoT LLM.	briefing	LLM or fallback.

Automated evaluation

Sentiment agreement (VADER vs LR): 92.0% — VADER and LR agree on 46/50 posts (92%). Use manually labelled gold set for formal accuracy in your report.

Briefing judge score: 70/100 (error_fallback)

LLM judge unavailable; use manual rubric in report.

11. Ethics, limitations & references

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- Public Reddit text only; no private user data.
- Crisis scores are decision-support estimates, not legal or PR advice.
- Sentiment models can misclassify sarcasm and domain-specific language.
- For quantitative evaluation on labelled data, extend `backend/evaluation.py`.
- Full NLP technique reference: `docs/NLP_TECHNIQUES.md`

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